OUTPUT 1:

Dataset loaded successfully

Country Name Country Code Year R&D_Expenditure_%GDP \

0 Australia AUS 1996 1.66218

1 Belgium BEL 1996 1.74299

2 Bulgaria BGR 1996 0.41826

3 Canada CAN 1996 1.60682

4 Switzerland CHE 1996 2.37649

Researchers_per_Million Patent_Applications

0 3353.70530 1797.0

1 2477.43906 776.0

2 1771.85422 316.0

3 3055.67247 2583.0

4 3130.03604 2504.0

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1637 entries, 0 to 1636

Data columns (total 6 columns):

Column Non-Null Count Dtype

--- ------ -------------- -----

0 Country Name 1637 non-null object

1 Country Code 1637 non-null object

2 Year 1637 non-null int64

3 R&D_Expenditure_%GDP 1637 non-null float64

4 Researchers_per_Million 1637 non-null float64

5 Patent_Applications 1637 non-null float64

dtypes: float64(3), int64(1), object(2)

memory usage: 76.9+ KB

None

Summary Statistics:

Year R&D_Expenditure_%GDP

Researchers_per_Million \ count 1637.000000 1637.000000

1637.000000

mean 2009.877214 1.209235 2301.454978

std 7.170740 0.951421 1944.663588

min 1996.000000 0.011040 13.283560

25% 2004.000000 0.438940 635.076300

50% 2010.000000 0.896260 1789.953560

75% 2016.000000 1.899060 3558.204670

max 2021.000000 4.909880 9071.450300

Patent_Applications

count 1.637000e+03

mean 6.563170e+04

std 2.235141e+05

min 1.000000e+00

25% 1.340000e+02

50% 7.890000e+02

75% 4.280000e+03

max 2.386300e+06

Missing Values:

Country Name 0

Country Code 0

Year 0

R&D_Expenditure_%GDP 0

Researchers_per_Million 0

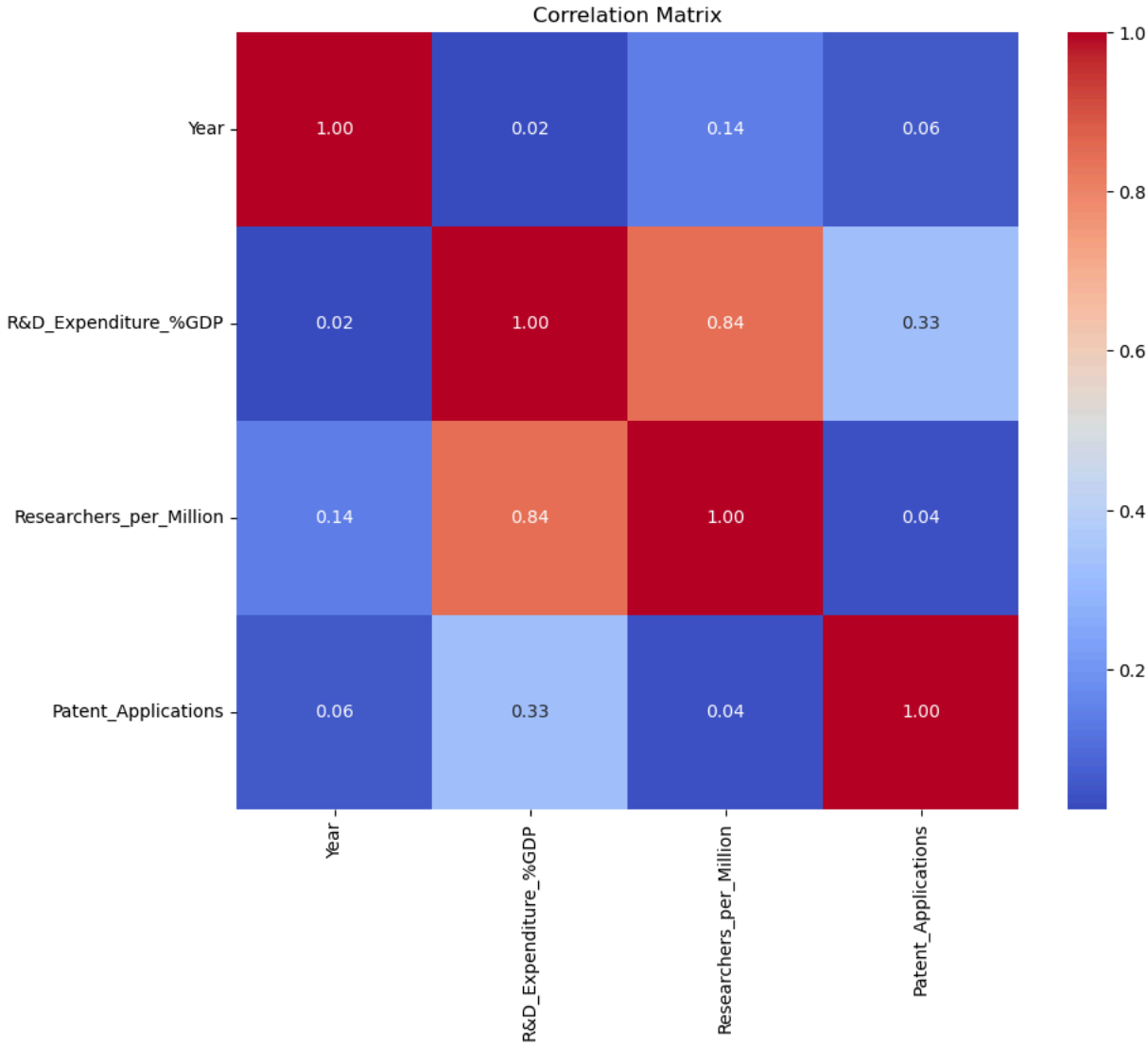
Patent_Applications 0

dtype: int64

The above attached output show that the Dataset loaded successfully

- There are 1637 records present in dataset
- Across 6 columns
- No missing values
- Correct data types (float, int, object)
- Since there are no missing values, the model does not require imputation or data cleaning steps before training. This ensures that model performance is not artificially
- affected by missing data handling techniques. Correct data types (float, int) also ensure mathematical operations and regression algorithms function properly.
- This confirms the dataset is clean and ready for modeling without preprocessing issues.
- The output also shows Mean, standard deviation, min, max
  Distribution of R&D expenditure, researchers per million, and patent applications
  This output provides us with statistical understanding of variable scale and distribution before modeling.

OUTPUT 2/CORRELATION MATRIX HEATMAP:



From the correlation matrix heatmap, we could observe that:

Correlation measures the strength and direction of linear relationships between variables. Values close to +1 indicate strong positive relationships, values near 0 indicate weak relationships, and values close to -1 indicate strong negative relationships.

R&D vs Researchers = 0.84 (Strong positive)
R&D vs Patent Applications = 0.33 (Moderate)
Researchers vs Patent Applications = 0.04 (Weak linear)

OUTPUT 3/MULTICOLLINEARITY CHECK/VIF:

```
Variance Inflation Factor (VIF):
                   Feature       VIF
0     R&D_Expenditure_%GDP  3.402784
1  Researchers_per_Million  3.402784
```

From the above output we observe that:

R&D Expenditure is equal to 3.40

Multicollinearity occurs when independent variables are highly correlated with each other. High multicollinearity can distort regression coefficients and reduce model interpretability. VIF (Variance Inflation Factor) quantifies how much variance of a coefficient increases due to correlation among predictors
Researchers per Million = 3.40 Since VIF < 5, multicollinearity is moderate but acceptable. Independent variables are not excessively correlated.

OUTPUT 4/PERFORMANCE COMPARISION ANALYSIS:

SDG 9 Impact Analysis Report

R&D_Expenditure_%GDP importance: 0.323

Researchers_per_Million importance: 0.677

Model explains 77.51% variation in patent output.

(The Random Forest model explains 77.51% of the variation in patent applications. This means that approximately 77.5% of the changes in patent output across countries can be predicted using R&D expenditure and number of researchers.)

Linear Regression Model Performance:

R^2 Score: 0.345

RMSE: 165,027

Random Forest Model Performance:

R^2 Score: 0.775

RMSE: 96,740

(RMSE (Root Mean Square Error) measures the average magnitude of prediction error. A lower RMSE indicates that predicted patent values are closer to actual values. Since patent counts vary widely, RMSE appears large in absolute value, but must be interpreted relative to the scale of patent data.)

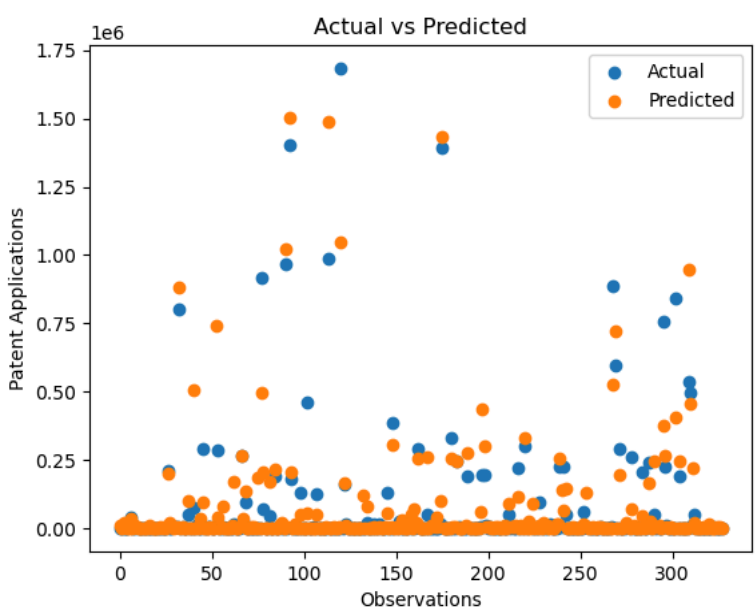
Model Comparison

Linear Regression R^2: 0.345

Random Forest R^2: 0.775

From this output we can conclude that Random Forest significantly outperforms Linear Regression, indicating non-linear relationships in the dataset.

OUTPUT 4/ACTUAL VS PREDICTED PLOT:



The Above Screenshot shows Actual vs Predicted Plots, where blue dot's indicate actual values and orange dots indicate predicted values We can also note that Random Forest predictions were closer to actual values

Extreme patent values are harder to predict because the dataset is highly skewed, with a small number of countries having extremely high patent counts. Machine learning models generally perform better on dense data regions than on sparse extreme cases

Slight deviation at extreme patent values.
TO CHECK DIFFERNECE IN VALUES OF THE ABOVE OUTPUTS, WE CHANGED VALUES OF SOME INPUTS:

```python
# TRAIN MODEL
def train_model_lr(X, y):

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, random_state=42
    )

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    return model, X_test, y_test, y_pred
```

0.0s

```python
#Random Forest Model
def train_model_rf(X, y, n_estimators=500, random_state=42):

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, random_state=random_state
    )

    model = RandomForestRegressor(
        n_estimators=n_estimators,
        random_state=random_state
    )

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    return model, X_test, y_test, y_pred
```

Previously the test_size in train,test was kept at 0.2, which is now changed to 0.3, also the value for n_estimators was kept at 200, which is further changed to 500, the key differences we note are:
Increasing the test size reduces the amount of training data available for the model to learn patterns, which can slightly reduce model accuracy.

SDG 9 Impact Analysis Report

R&D_Expenditure_%GDP importance: 0.327

Researchers_per_Million importance: 0.673

Model explains 75.58% variation in patent output.

Model Performance:

R^2 Score: 0.311

RMSE: 158786.698

Model Performance:

R^2 Score: 0.756

RMSE: 94508.352

Model Comparison

Linear Regression R^2: 0.311

Random Forest R^2: 0.756

From the above output, we can note that compared to previous performance metrics, the updated performance metrics give:

Linear Regression:

- $R^2 = 0.311$
- RMSE = 158,786

Random Forest:

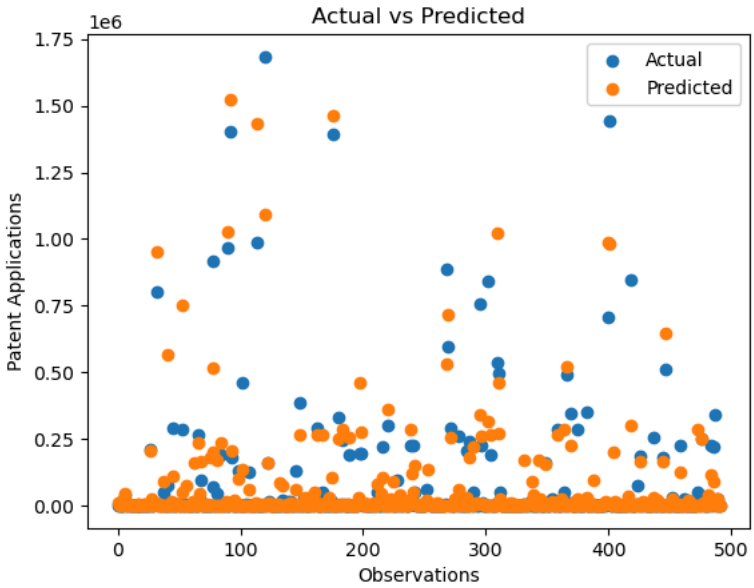
- $R^2 = 0.756$
- RMSE = 94,508

Performance Comparison Analysis

Observations:

1. Increasing test size reduced training data, causing a slight decrease in R^2 for both models.
2. A slight decrease in R^2 is expected when training data is reduced, as the model has fewer examples to learn complex patterns
3. Increasing the number of trees improved model stability and slightly reduced RMSE.
4. Random Forest continues to outperform Linear Regression significantly.
5. The dataset exhibits non-linear behavior, which Random Forest handles better.

Updated Actual vs Predicted Plot(With test_size = 0.3 and n_estimators = 500):



Changes in scatter plot for test_size=0.3 and n_estimators = 500, comapred with scatter plot of test_size = 0.2 and n_estimators = 200:

Increasing the test size from 0.2 to 0.3 increased the number of test observations, resulting in a denser scatter plot. The overall pattern remains similar, indicating model stability. However, a slight increase in prediction dispersion is observed, consistent with the small decrease in R^2 score.

FINAL CONCLUSION :
Overall, the analysis demonstrates that patent generation is influenced by non-linear relationships between R&D expenditure and researcher density. Multicollinearity is within acceptable limits, and Random Forest consistently outperforms Linear Regression. The model remains stable under parameter modifications, confirming robustness of the predictive approach for SDG 9 analysis.